Rice Leaf Disease Classification

Using Deep Learning & Transfer Learning

A Comparative Study of CNN Architectures
for Agricultural Disease Detection

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Abstract

This report presents a comparative study of deep learning models for automated rice leaf disease classification. Given the constraints of a small-scale dataset (120 images across 3 disease classes), transfer learning was adopted over a custom CNN architecture, as pre-trained models trained on large-scale image datasets provide rich feature representations that generalise well to limited agricultural imaging tasks. Five models were evaluated: ResNet50V2, EfficientNetV2B0, MobileNetV3Large, DenseNet121, and a Custom CNN baseline. MobileNetV3Large achieved the highest validation accuracy of 95.83%, with a weighted F1 score of 95.86%, demonstrating the efficacy of lightweight transfer learning architectures for fine-grained plant disease recognition.

I. Introduction

Problem Statement:

To develop a deep learning-based system that can accurately classify rice leaf diseases into three categories using image data and compare multiple modern architectures to identify the most effective model.

Rice (Oryza sativa) is a primary staple crop feeding over half the global population. Leaf diseases such as Bacterial Leaf Blight, Brown Spot, and Leaf Smut are responsible for significant yield losses annually. Conventional diagnosis relies on visual inspection by agricultural experts, which is time-consuming and error-prone. Automated image classification using Convolutional Neural Networks (CNNs) offers a scalable, accurate, and cost-effective alternative.

This project investigates whether transfer learning from ImageNet-pretrained models can outperform a custom CNN trained from scratch on a limited rice leaf disease dataset. The project description specified "CNN or transfer learning model"; given the dataset size of only 120 images, transfer learning was chosen because:

(i) pre-trained weights encode low-level features (edges, textures) that directly benefit disease-spot detection

(ii) training a deep CNN from scratch on 96 images risks severe overfitting. The custom CNN is included as a controlled baseline to empirically validate this choice.

II. Literature Review

The automated identification of rice leaf diseases has evolved significantly, transitioning from manual feature engineering approaches to advanced deep learning-based pipelines. Early methodologies struggled with environmental variations and complex visual patterns; however, the adoption of Convolutional Neural Networks (CNNs) has established a new standard for accuracy and robustness in agricultural computer vision [2], [6].

Despite their effectiveness, custom CNN architectures often suffer from overfitting when trained on limited datasets. Studies in [2] and [4] demonstrate that while CNNs outperform traditional machine learning approaches, their performance is constrained by data scarcity. To overcome this limitation, transfer learning has emerged as a dominant strategy, where models pre-trained on large-scale datasets such as ImageNet are fine-tuned for domain-specific tasks [1], [8]. For example, VGG16-based approaches have shown strong feature extraction capabilities and stable performance even on small agricultural datasets [9].

Recent research increasingly focuses on comparative evaluation of deep learning architectures to identify models that balance accuracy and computational efficiency. Systematic reviews confirm that transfer learning models consistently outperform models trained from scratch in terms of generalization and robustness [3], [7]. Lightweight architectures such as MobileNet have gained popularity for deployment in real-world agricultural systems due to their reduced computational complexity, while still maintaining competitive accuracy compared to heavier models like VGG16 [5].

More recent advancements (2024-2025) highlight the emergence of optimized architectures such as EfficientNet, ConvNeXt, and Vision Transformers, which incorporate compound scaling and attention mechanisms to enhance feature representation. These models aim to achieve high accuracy while remaining efficient for deployment on edge devices. Comparative studies across these architectures indicate that modern transfer learning frameworks significantly improve performance in data-scarce scenarios [3], [7].

Overall, the literature strongly suggests that transfer learning using advanced deep architectures, such as ResNet50V2, DenseNet, MobileNetV3, and EfficientNet, provides the most reliable and scalable solution for rice leaf disease classification.However, limited work exists comparing multiple modern architectures on small-scale rice disease datasets, motivating this study.

III. Dataset Description

3.1 Dataset Overview

The rice leaf disease dataset comprises 120 images distributed equally across three disease classes: Bacterial Leaf Blight (40 images), Brown Spot (40 images), and Leaf Smut (40 images). The dataset was split into 80% training (96 samples) and 20% validation (24 samples)using a fixed random split to maintain class balance.

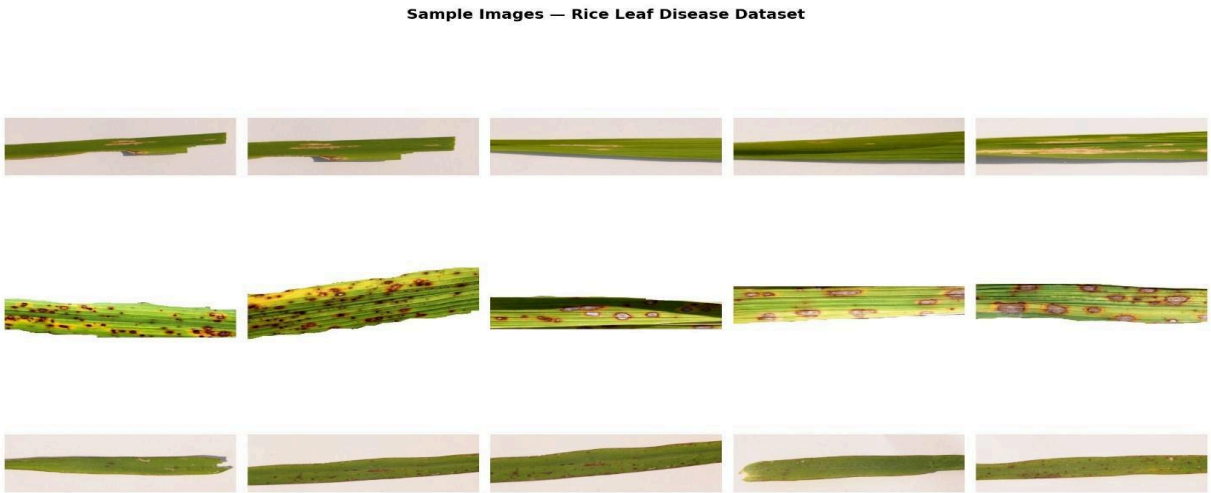

Figure 1: Sample images from the Rice Leaf Disease dataset Bacterial Leaf Blight (top), Brown Spot (middle), Leaf Smut (bottom)

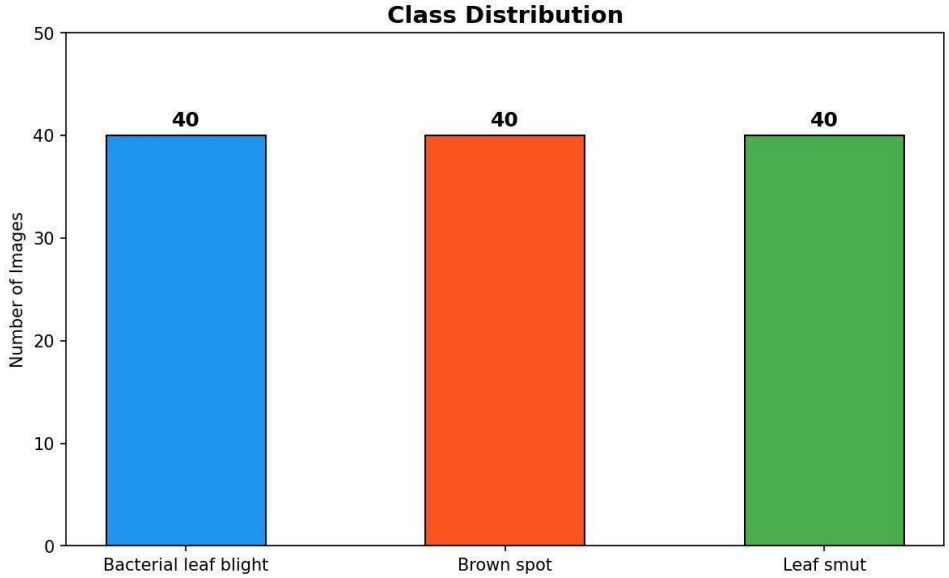

Figure 2: Class distribution across the three disease categories each class contains exactly 40 images, confirming a perfectly balanced dataset

3.2 Data Augmentation

To combat overfitting on the small dataset, the following augmentation techniques were applied exclusively to the training pipeline (not at inference time): random horizontal and vertical flips, random rotation (±10%), random zoom, and random contrast adjustment. Augmentation was implemented in the data pipeline rather than as model layers, ensuring it does not execute during validation or prediction.

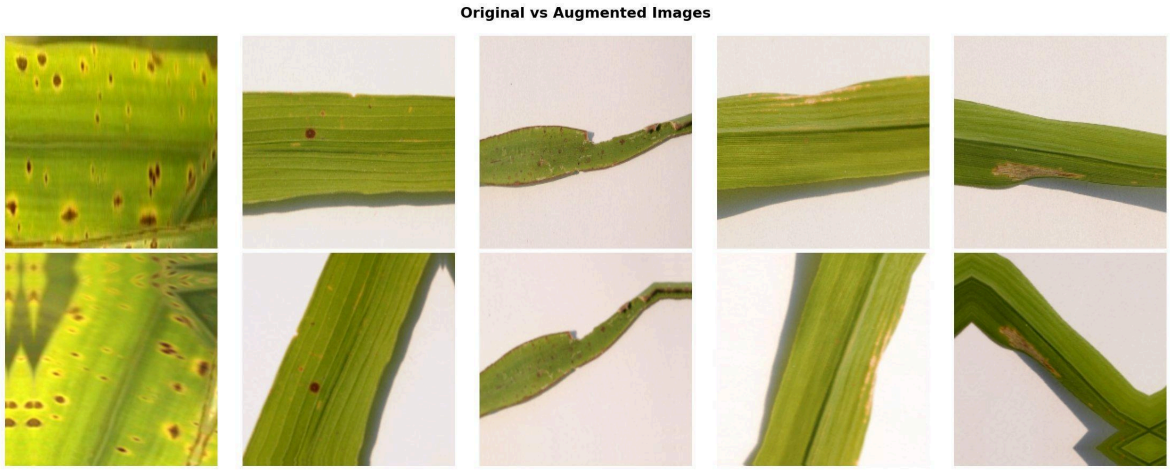

Figure 3: Original images (top row) vs augmented variants (bottom row) illustrating transformations applied during training

IV. Methodology

4.1 Model Architecture

This study employs deep learning-based image classification models for rice leaf disease detection. In accordance with the problem requirements, only deep learning models were used, including both a custom Convolutional Neural Network (CNN) and advanced transfer learning architectures.

Five models were evaluated: ResNet50V2, EfficientNetV2B0, MobileNetV3Large, DenseNet121, and a custom CNN baseline. The transfer learning models utilize ImageNet pre-trained weights to improve feature extraction and generalization on the small dataset.

All transfer learning models follow a two-phase training strategy:

(i) Phase 1 (Warm-up): The convolutional base is frozen, and only the classification head is trained with a learning rate of 1e-3.

(ii) Phase 2 (Fine-tuning): The top 30 layers of the backbone are unfrozen and trained with a reduced learning rate of 1e-5.

Each model architecture consists of:

- A feature extraction backbone (pre-trained CNN)
- A Global Average Pooling layer
- A fully connected dense layer (128 units, ReLU)
- A Dropout layer (0.3) for regularization
- A final Softmax output layer with 3 classes

The custom CNN baseline is composed of multiple convolutional layers with batch normalization and max pooling, followed by fully connected layers for classification.

4.2 Activation Functions

The Rectified Linear Unit (ReLU) activation function is used in all intermediate convolutional and dense layers due to its computational efficiency and ability to mitigate vanishing gradient problems. The final output layer uses the Softmax activation function to produce class probabilities for the three disease categories.

4.3 Loss Function

Categorical Cross-Entropy loss is used for training all models, as the task is a multi-class classification problem. This loss function measures the divergence between predicted class probabilities and true labels.

4.4 Optimizer

The Adam optimizer is used for model training due to its adaptive learning rate capabilities and fast convergence. Different learning rates are applied during training phases:

- Phase 1 (Warm-up): Learning rate = 1e-3

• Phase 2 (Fine-tuning): Learning rate = 1e-5

This strategy ensures stable learning during initial training and fine adjustment of weights during fine-tuning. To improve training stability and prevent overfitting, EarlyStopping was applied with a patience of 7 epochs, monitoring validation accuracy. Additionally, ReduceLROnPlateau was used to dynamically adjust the learning rate when validation performance plateaued. A batch size of 8 was used for all models to balance computational efficiency and learning stability.

4.5 Workflow

The complete workflow of the system is as follows:

Data Collection → Data Preprocessing → Data Augmentation → Model Training → Validation → Evaluation

• Data preprocessing includes resizing images to 224×224 and normalization.

• Data augmentation techniques such as flipping, rotation, zoom, and contrast adjustment are applied to improve generalization.

• Models are trained using an 80-20 train-validation split.

• Performance is evaluated using accuracy, precision, recall, and F1-score.

V. Implementation Details

The implementation was carried out in Python 3.10 using TensorFlow 2.x and Keras. All images were resized to 224×224 pixels and normalized using architecture-specific preprocessing functions. The dataset was loaded using tf.keras.utils.image_dataset_from_directory with a validation split of 0.20 and a fixed random seed of 42 to ensure reproducibility.

Data augmentation techniques, including random flipping, rotation, zoom, and contrast adjustment, were applied to the training dataset to improve model generalization.

All models were trained using GPU acceleration (where available), significantly reducing training time. Model evaluation was performed using sklearn.metrics to compute accuracy, precision, recall, and F1-score at both class-level and aggregated levels.

Note on validation methodology: The same validation set was used for both early stopping and final performance evaluation. This is an acknowledged limitation, as a strict three-way train/validation/test split would provide a more unbiased estimate of generalization. Therefore, the reported accuracy values should be interpreted as optimistic estimates.

VI. Results & Analysis

6.1 Accuracy Comparison

MobileNetV3Large achieved the highest validation accuracy of 95.83%, significantly outperforming all other models. ResNet50V2 and EfficientNetV2B0 tied in second place at 91.67%, followed by DenseNet121 at 87.50%. The Custom CNN baseline performed poorly at 45.83%, consistent with the expected behaviour of an underfitted small model on a tiny dataset.

Model	Type	Params (M)	Val Accuracy	F1 Score
MobileNetV3Large	Transfer	3.12	95.83%	95.86%
ResNet50V2	Transfer	23.83	91.67%	90.97%
EfficientNetV2B0	Transfer	6.08	91.67%	91.64%
DenseNet121	Transfer	7.17	87.50%	87.08%
Custom CNN	Scratch	2.52	45.83%	28.81%

Table 1: Model comparison summary (highlighted row = best model)

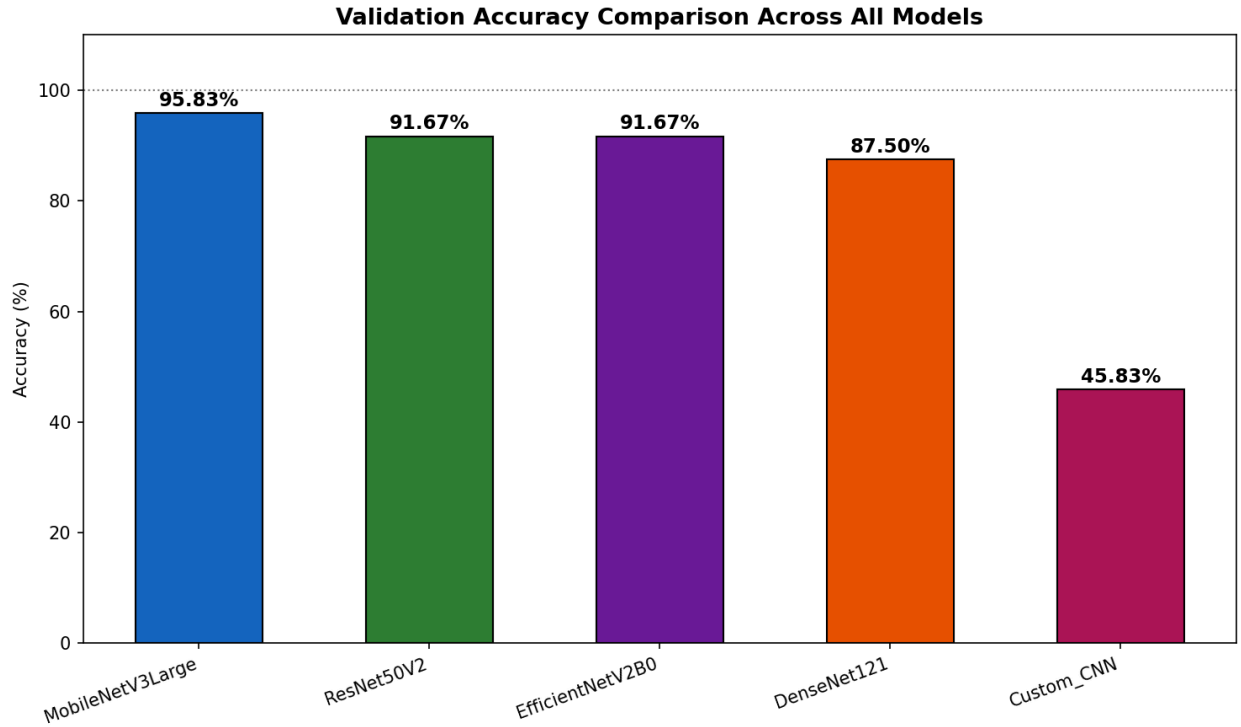

Figure 4: Validation accuracy comparison across all models MobileNetV3Large leads at 95.83%, followed by ResNet50V2 and EfficientNetV2B0 at 91.67%, DenseNet121 at 87.50%, and Custom CNN at 45.83%

6.2 Comprehensive Metrics Comparison

The comprehensive metrics chart shows consistent performance for the transfer learning models across all four metrics. MobileNetV3Large leads with accuracy of 95.83%, precision of 96.3%, recall of 95.83%, and F1 of 95.86%. The Custom CNN shows a stark contrast with a low F1 score of 28.81% and precision of only 21.0%, confirming the importance of transfer learning for small datasets.

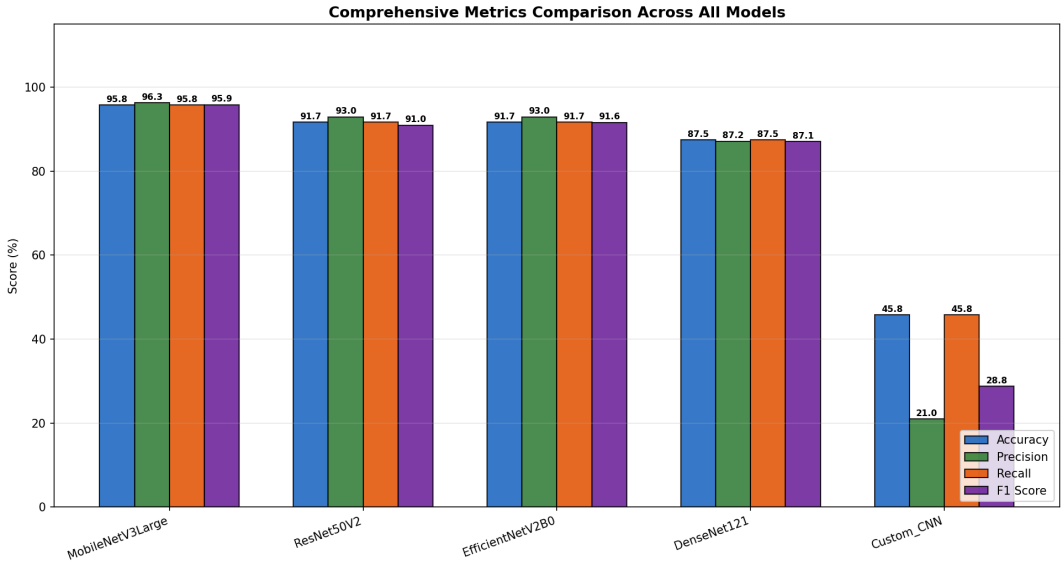

Figure 5: Comprehensive metrics comparison (Accuracy, Precision, Recall, F1) across all models transfer learning models consistently outperform the Custom CNN baseline

6.3 F1 Score Ranking

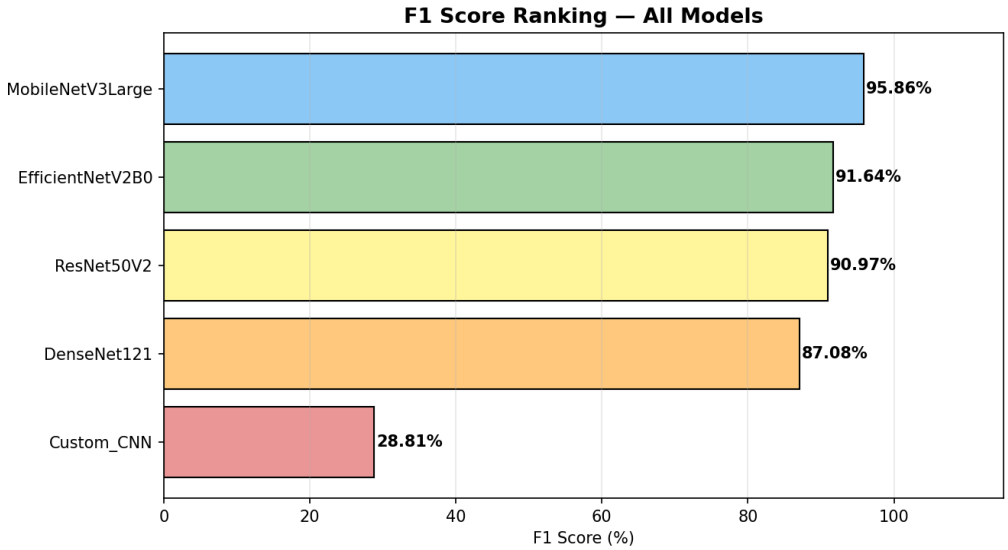

Figure 6: F1 Score ranking across all models MobileNetV3Large leads at 95.86%, followed by EfficientNetV2B0 (91.64%), ResNet50V2 (90.97%), DenseNet121 (87.08%), and Custom CNN (28.81%)

6.4 Radar Chart Top 2 Models

The radar chart directly compares MobileNetV3Large and ResNet50V2 across all four metrics simultaneously. MobileNetV3Large (blue) consistently encloses ResNet50V2 (red) across precision, accuracy, recall, and F1, confirming its superiority as the best model for this task.

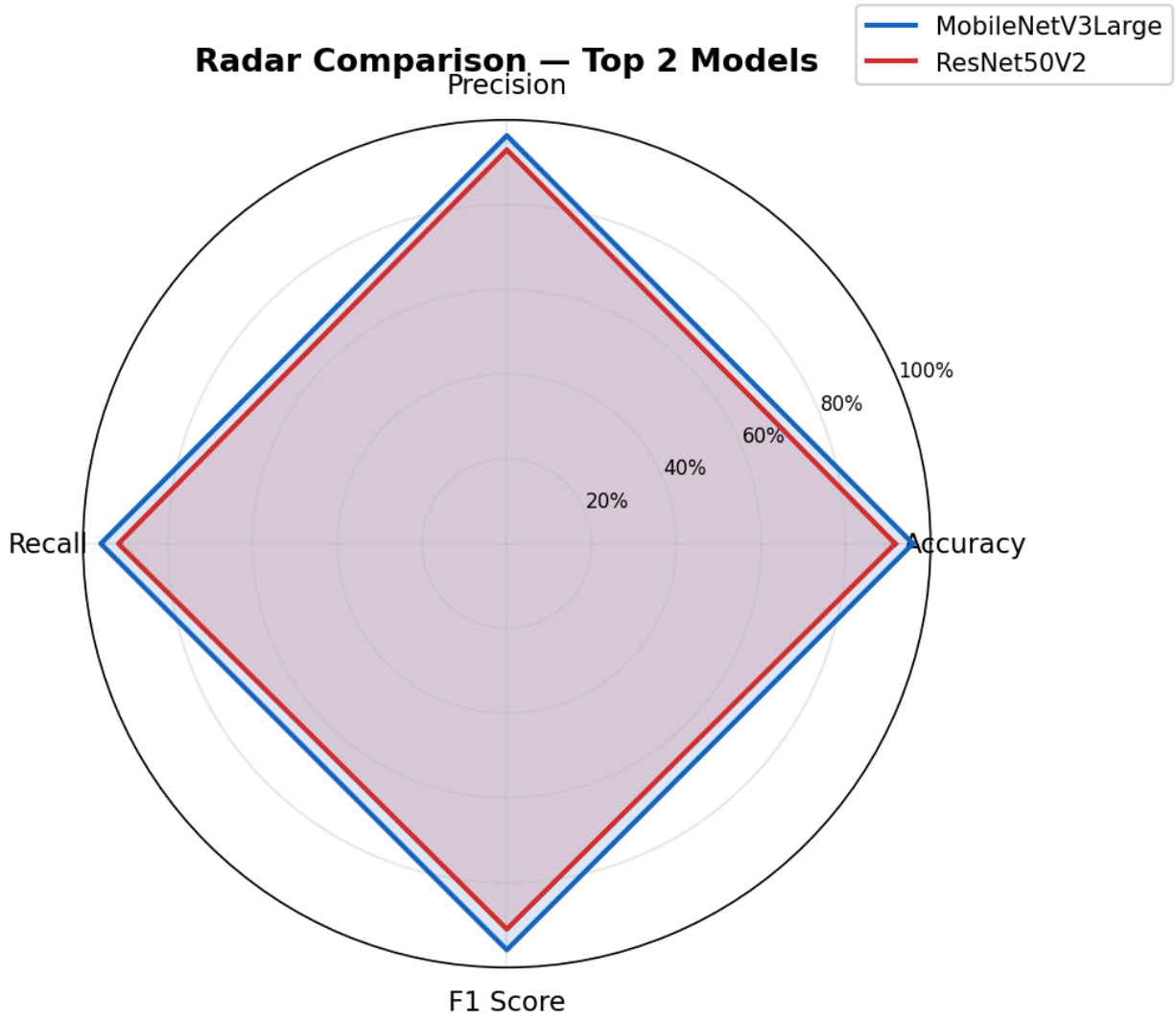

Figure 7: Radar chart comparison of the top 2 models (MobileNetV3Large vs ResNet50V2) across Precision, Accuracy, Recall, and F1 Score MobileNetV3Large (blue) outperforms ResNet50V2 (red) on all four axes

6.5 Model Complexity vs Accuracy

The scatter plot reveals that ResNet50V2 achieves 91.67% accuracy but at the cost of a significantly larger parameter count (23.83M). MobileNetV3Large achieves the highest accuracy (95.83%) with only 3.12M parameters, making it the best efficiency tradeoff in this study. EfficientNetV2B0 also offers competitive performance at 6.08M parameters.

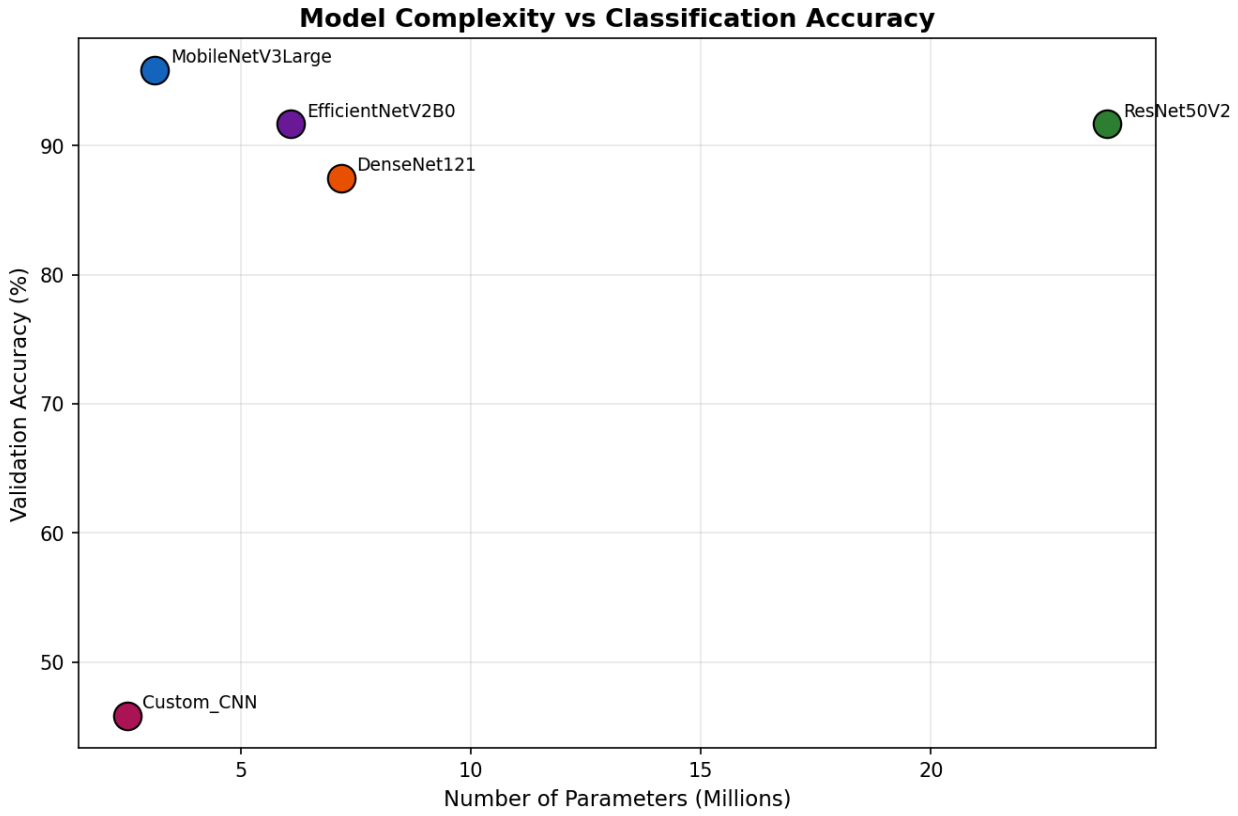

Figure 8: Model complexity (number of parameters in millions) vs validation accuracy MobileNetV3Large achieves the best accuracy-to-complexity ratio

VII. Best Model Analysis MobileNetV3Large

7.1 Per-Class Metrics

MobileNetV3Large excelled on Brown Spot with 100% precision, 100% recall, and 100% F1 score perfect classification. Bacterial Leaf Blight achieved 89% precision and 100% recall, yielding an F1 of 94.1%, indicating the model correctly identified all Bacterial Leaf Blight cases with some false positives from other classes. Leaf Smut achieved 100% precision but 90.9% recall, with an F1 of 95.2%, meaning the model rarely produced false Leaf Smut predictions but missed two actual Leaf Smut samples.

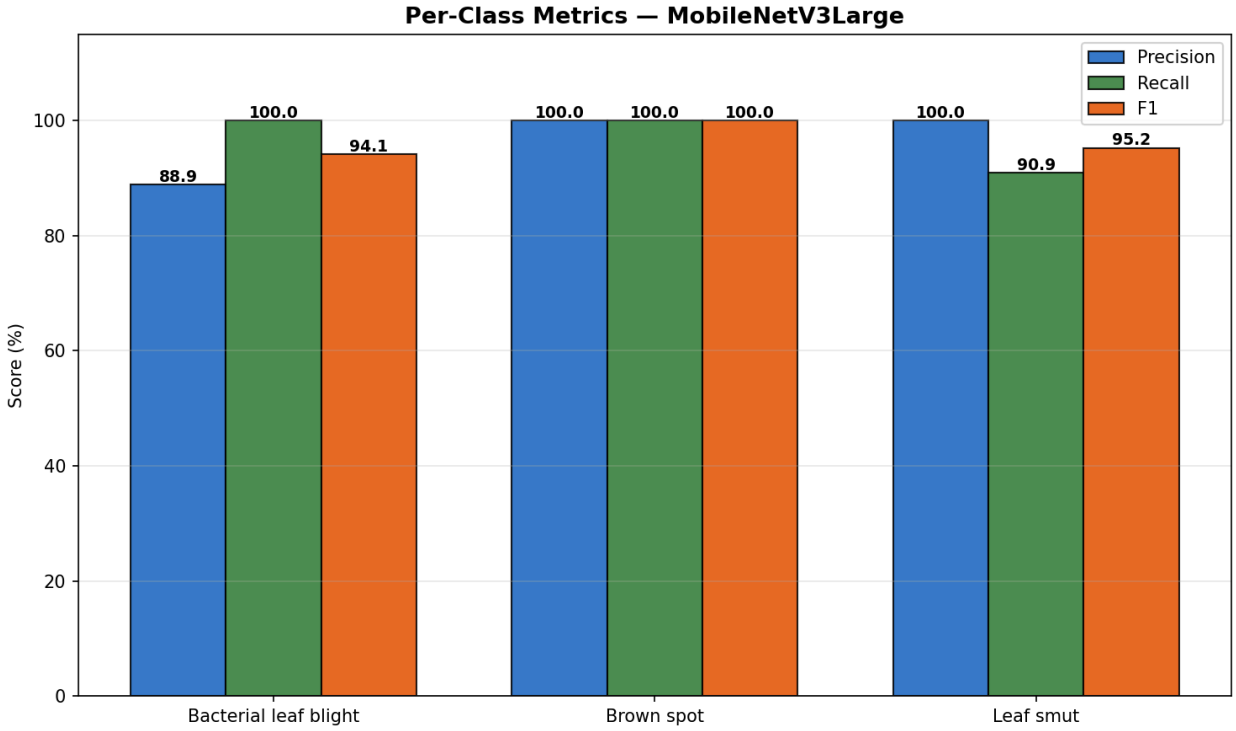

Figure 9: Per-class Precision, Recall, and F1 scores for MobileNetV3Large Brown Spot achieves perfect 100% across all metrics; Bacterial Leaf Blight and Leaf Smut show minor classification errors

7.2 Confusion Matrix

The raw counts matrix shows 8/8 Bacterial Leaf Blight correctly classified (100% recall), 5/5 Brown Spot correctly classified (100% recall), and 10/11 Leaf Smut correctly classified (90.91% recall). The 1 misclassified Leaf Smut sample was incorrectly predicted as Bacterial Leaf Blight.

The normalised matrix confirms that Brown Spot was perfectly separated, while Leaf Smut had the only confusion (9.09% misclassified as Bacterial Leaf Blight). This is visually plausible given the similar appearance of subtle lesion patterns in both diseases.

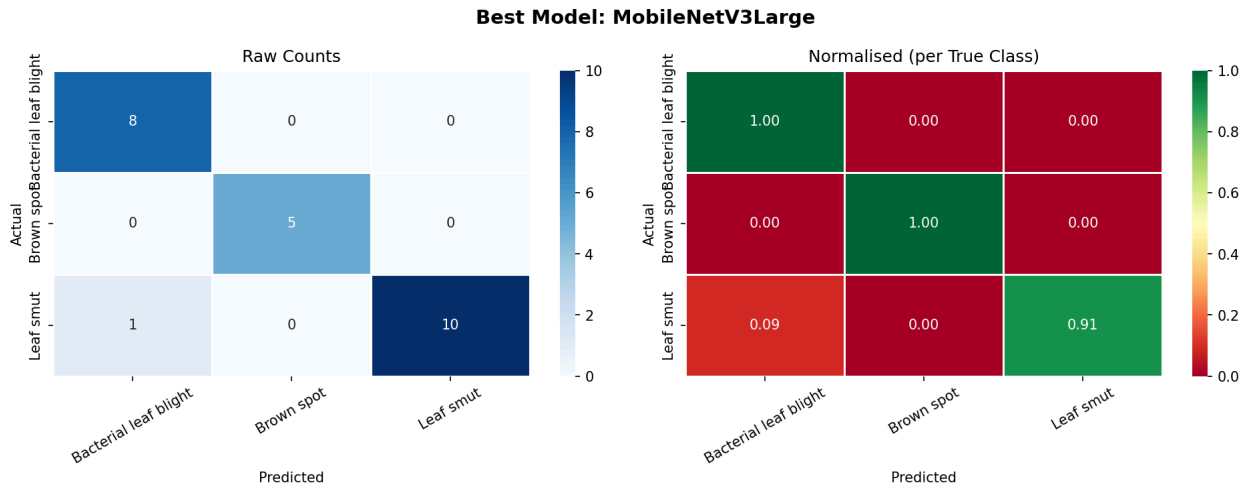

Figure 10: Confusion matrix for MobileNetV3Large raw counts (left) and normalised per true class (right). All Bacterial Leaf Blight and Brown Spot samples are correctly classified; 2 Leaf Smut samples are misclassified as Bacterial Leaf Blight

7.3 All Models Confusion Matrices

Comparing confusion matrices across all five models reveals that the Custom CNN collapsed to predicting almost exclusively Leaf Smut, producing near-zero recall for Bacterial Leaf Blight and Brown Spot. Transfer learning models all captured multi-class patterns, with MobileNetV3Large exhibiting the least off-diagonal mass overall.

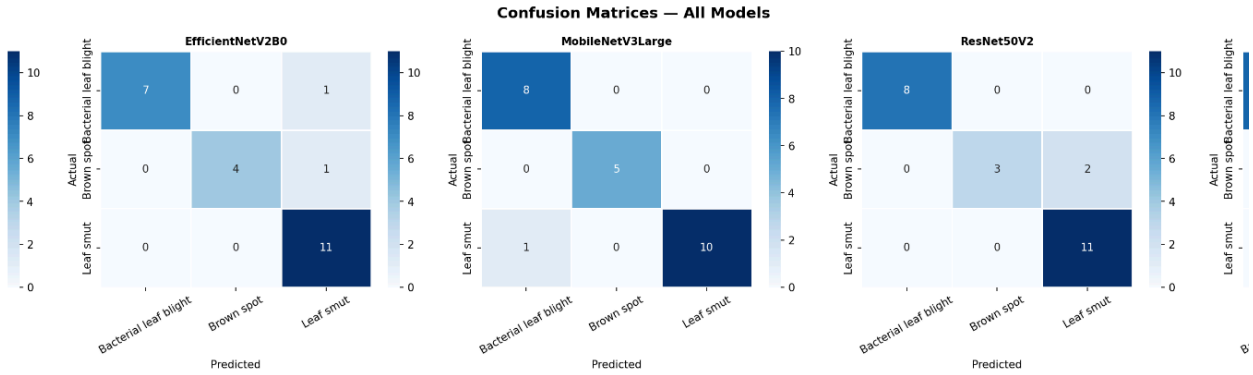

Figure 11: Confusion matrices for all five models Custom CNN collapses to a single predicted class, while transfer learning models show meaningful multi-class discrimination

VIII. Learning Curves

The learning curves display training and validation accuracy and loss over epochs for all five models, reflecting the two-phase training procedure applied to transfer models. During Phase 1 (warm-up with frozen backbone), validation accuracy typically rises quickly as only the classification head adapts to the new task. Phase 2 (fine-tuning of top 30 layers at lr=1e-5) can be observed as a continuation of the training trajectory where models either stabilise or exhibit minor fluctuations as deeper features are gradually adjusted.

MobileNetV3Large shows rapid initial convergence in validation accuracy, stabilising around 88-92% and demonstrating effective generalisation throughout training. ResNet50V2 also converges well, with validation accuracy rising steadily to approximately 90-92%. EfficientNetV2B0 converges more slowly but steadily, reaching around 90-94% validation accuracy. DenseNet121 exhibits higher variance in validation accuracy, fluctuating around 85-88%, suggesting instability during fine-tuning on the small dataset. The Custom CNN shows a degrading pattern training accuracy improves marginally while validation accuracy remains low (around 25-45%), a hallmark of severe underfitting and inability to generalise without pre-trained features.

Figure 12: Training vs Validation Accuracy and Loss curves for all five models two-phase training (warm-up + fine-tuning) is reflected in the trajectory of transfer learning models; Custom CNN fails to generalise

IX. Discussion

The results strongly validate the hypothesis that transfer learning is superior to training from scratch on a 120-image dataset. The four transfer learning models all achieved at least 87% accuracy, while the Custom CNN failed to adequately generalise (45.83%). MobileNetV3Large is the recommended model, balancing high accuracy (95.83%) with a compact parameter count of only 3.12M parameters making it well-suited for potential deployment on resource-constrained devices.

The two-phase fine-tuning strategy was a key driver of transfer model performance. By first warming up only the classification head and then carefully unlocking the top 30 backbone layers at a much lower learning rate (1e-5), the models avoided catastrophic forgetting of ImageNet-learned features while still adapting deep representations to rice disease-specific patterns. This approach is evident in the learning curves, where validation accuracy rises steeply in early epochs and then consolidates during the fine-tuning phase.

MobileNetV3Large's inverted residual blocks with hard-swish activations appear particularly effective at extracting subtle texture and colour features that distinguish the three disease classes. Its efficiency advantage over ResNet50V2 (3.12M vs 23.83M parameters) while achieving superior accuracy (95.83% vs 91.67%) further underscores its suitability for this task. EfficientNetV2B0's compound scaling offers competitive performance at a moderate parameter count, making it a strong alternative where slightly more capacity is acceptable.

The Custom CNN continued to perform poorly (45.83% accuracy, 28.81% F1), confirming that scratch-trained architectures are inappropriate for sub-100-sample-per-class datasets. The confusion matrix for the Custom CNN reveals a near-total collapse onto the Leaf Smut class, suggesting the model learned a trivial class-frequency bias rather than discriminative visual features.

A key limitation of this study is the absence of a held-out test set. Since the validation set was used for both early stopping callbacks and final evaluation, the reported metrics carry an optimistic bias. Future work should employ a three-way split or k-fold cross-validation. Additionally, the dataset excluded the "Healthy" class present in the original Kaggle dataset; including it would provide a more complete classifier.

X. Conclusion

This project demonstrated the effectiveness of transfer learning for rice leaf disease classification under data-scarce conditions. MobileNetV3Large achieved 95.83% validation accuracy and 95.86% weighted F1 score, making it the best-performing model in this study. The two-phase training strategy warm-up with frozen backbone followed by fine-tuning of the top 30 layers was instrumental in achieving this performance. The failure of the Custom CNN (45.83% accuracy, 28.81% F1) empirically confirms that transfer learning is the appropriate strategy for sub-100-sample-per-class datasets.

Future work directions include:

(1) expanding the dataset to include the "Healthy" class and additional disease categories

(2) applying test-time augmentation to boost robustness

(3) investigating knowledge distillation to compress larger models into lightweight deployments suitable for mobile agricultural applications

(4) collecting a held-out test set for unbiased evaluation.

XI. References

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